

Lecture 10: Geometry of Linear Programming Problems

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10.1 Matrix Equations and Constraints

Last time we ended with the matrix version of the linear programming problem:

Find the vector $\mathbf{x} \in \mathbb{R}^n$ that will

$$\text{minimize } z = \mathbf{c}^\top \mathbf{x} \text{ subject to the restrictions} \quad (10.1)$$

$$\left. \begin{array}{l} \mathbf{Ax} \leq \mathbf{b} \\ \mathbf{x} \geq \mathbf{0} \end{array} \right\} \quad (10.2)$$

When we say that $\mathbf{v} \leq \mathbf{w}$ for two vectors $\mathbf{v}$ and $\mathbf{w}$, we mean that each entry of $\mathbf{v}$ is less than or equal to the corresponding entry of $\mathbf{w}$. So if $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, meaning they both have n entries in them, we see that $\mathbf{v} \leq \mathbf{w}$ means $v_i \leq w_i$ for all $i = 1, 2, \dots, n$.

Example 10.1 Find the vector $\mathbf{x}$ in $\mathbb{R}^2$ that will

$$\begin{aligned} \text{maximize } z &= \begin{bmatrix} 120 & 100 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \text{ subject to} \\ &\left. \begin{array}{l} \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 8 \\ 15 \end{bmatrix} \\ \begin{bmatrix} x \\ y \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array} \right\} \end{aligned}$$

Definition 10.2 A vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfying the constraints of linear programming problem is called a feasible solution to the problem. A feasible solution that optimizes (either maximizes or minimizes) the objective function of a linear programming problem is called an optimal solution.

Example 10.3 Consider the linear programming problem above, where the constraints are

$$\begin{aligned} \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &\leq \begin{bmatrix} 8 \\ 15 \end{bmatrix} \\ \begin{bmatrix} x \\ y \end{bmatrix} &\geq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

The vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

are all feasible solutions because

$$\begin{aligned} \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \mathbf{x}_1 &= \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \end{bmatrix} \leq \begin{bmatrix} 8 \\ 15 \end{bmatrix} & \text{and } \mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \mathbf{x}_2 &= \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 13 \end{bmatrix} \leq \begin{bmatrix} 8 \\ 15 \end{bmatrix} & \text{and } \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \mathbf{x}_3 &= \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \end{bmatrix} \leq \begin{bmatrix} 8 \\ 15 \end{bmatrix} & \text{and } \mathbf{x}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

Similarly, we can see that

$$\mathbf{x}_4 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{x}_5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

are not feasible solutions because they violate the linear constraint, i.e.

$$\begin{aligned} \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \mathbf{x}_4 &= \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 18 \end{bmatrix} \geq \begin{bmatrix} 8 \\ 15 \end{bmatrix} \\ \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \mathbf{x}_5 &= \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 16 \end{bmatrix} \geq \begin{bmatrix} 8 \\ 15 \end{bmatrix} \end{aligned}$$

and

$$\mathbf{x}_6 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

violates the nonnegativity condition, since one of the entries are negative.

10.2 Geometry of Linear Programming problems

Let's look at a single constraint (let's say the i^{th} one) of a linear programming problem in standard form:

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \leq b_i,$$

can be written as

$$\mathbf{a}^\top \mathbf{x} \leq b_i,$$

where $\mathbf{a}^\top = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$

Definition 10.4 The set of points $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ that satisfy this constraint is called a closed half-space. If the inequality is reversed, the set of points $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ satisfying

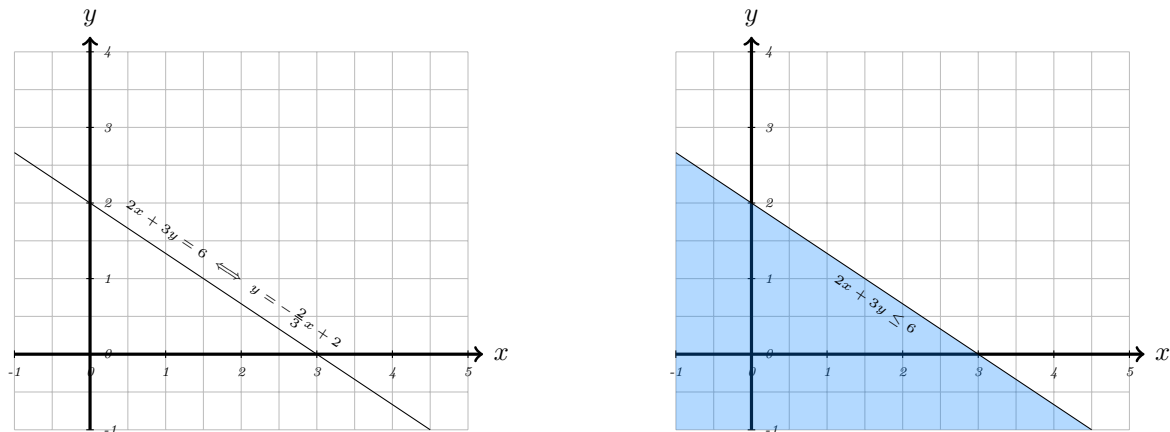
$$\mathbf{a}^\top \mathbf{x} \geq b_i$$

is also called a closed half-space.

Example 10.5 Consider the constraint $2x + 3y \leq 6$ and the closed half-space

$$H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \mid \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq 6 \right\}.$$

To visualize what this looks like, we have to graph the line and see what side of the space the inequality satisfies. First we see that $2x + 3y = 6$ can be rewritten as $y = -\frac{2}{3}x + 2$, which means that when $x = 0, y = 2$, and when $y = 0$ happens when $x = 3$. Connecting the two points, we have the line. Now this is the equality, but we want to find all the points that satisfy < 6 as well. So we pick any point to see if it satisfies this inequality. A good point to pick is $(0,0)$, where we see $2(0) + 3(0) \leq 6$, so since this is correct, all points on that side of the line work. This means the closed half space is visualized below



Example 10.6 Consider the constraint in 3 dimensions $4x + 2y + 5z \leq 20$ and the closed half-space

$$H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid \begin{bmatrix} 4 & 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \leq 20 \right\}.$$

Just like we did before, we set one of the variables equal to zero and then plot the line in that plane. For example if $z = 0$, then we have $4x + 2y \leq 20$. This means it has intercept points of $(x, y) = (0, 10)$ and $(5, 0)$. So the constraint connects $(5, 0, 0)$ and $(0, 10, 0)$ in the xy -plane.

Now setting $y = 0$, we have $4x + 5z \leq 20$, which connects $(5, 0, 0)$ and $(0, 0, 4)$. Lastly, we have $x = 0$, which means $2y + 5z \leq 20$, which connects $(0, 10, 0)$ and $(0, 0, 4)$. Since we see that the point $(0, 0, 0)$ is not on the plane, we use that as the test point. We see that $4(0) + 2(0) + 5(0) \leq 20$, so that means everything on the same side as the origin is a feasible point. This means we get the following structure.

Definition 10.7 A typical constraint of a linear programming problem in canonical form has the equation $\mathbf{a}^\top \mathbf{x} = b$. Just like with the 3D example we did the first day of the semester, the solution to a two dimensional problem is a line, the three dimensional problem is a plane, and the solution to $\mathbf{a}^\top \mathbf{x} = b$ is called a hyperplane. If the equation was an inequality, namely $\mathbf{a}^\top \mathbf{x} \leq b$, then the set of points satisfying the inequality would be the closed half space.

Remark 10.8 A hyperplane is the **boundary** of a closed half space. Intuitively, it consists of the points that are in the half-space, but on its edge.

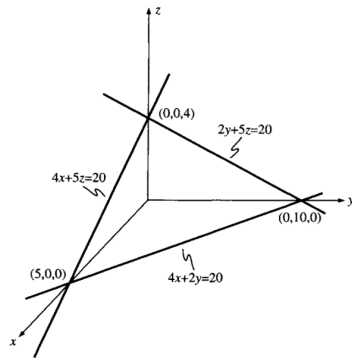


Figure 10.1:

However, there are very few instances where you will have just one constraint. In fact, you will probably have more constraints than variables most of the time! This means we need to figure out a way to combine these constraints to find the set of points that satisfy them all.

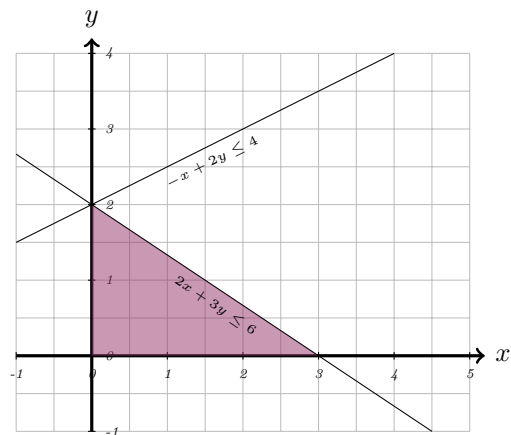
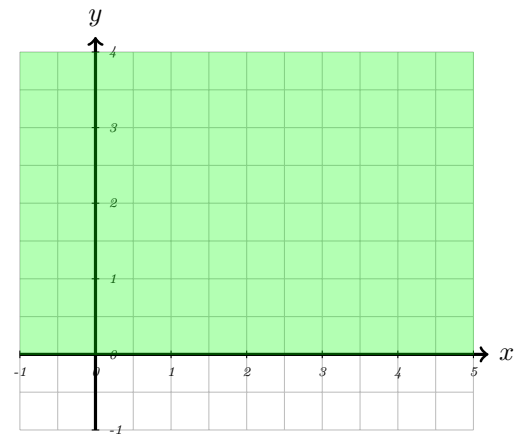
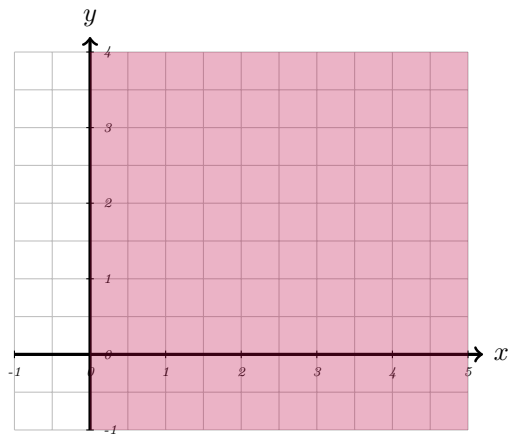
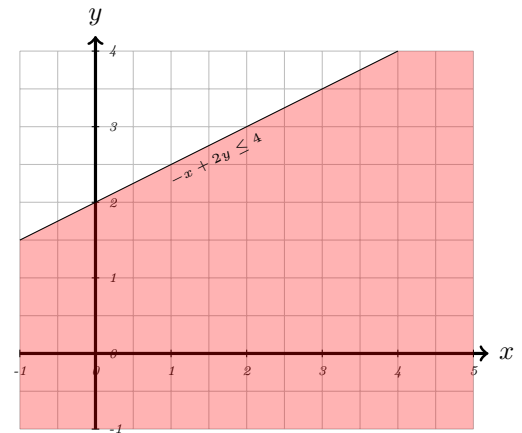
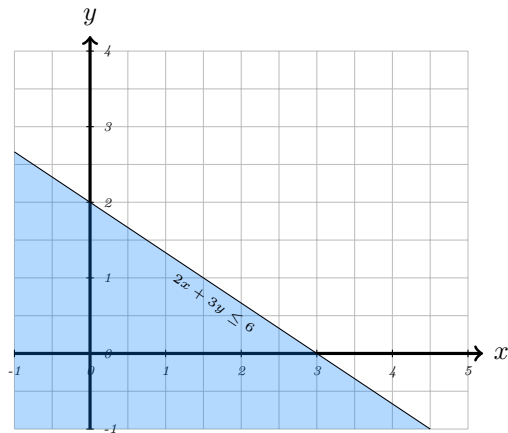
Example 10.9 *Sketch the set of all feasible solutions satisfying the set of inequalities:*

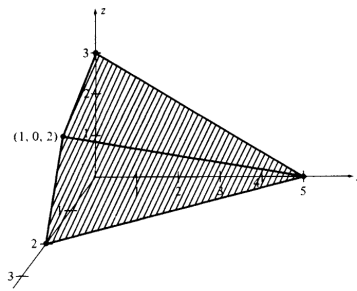
$$2x + 3y \leq 6$$

$$-x + 2y \leq 4$$

$$x \geq 0$$

$$y \geq 0$$





Example 10.10 Sketch the set of all feasible solutions satisfying the set of inequalities:

$$5x + 3y + 5z \leq 15$$

$$10x + 4y + 5z \leq 20$$

$$x \geq 0$$

$$y \geq 0$$

$$z \geq 0$$